

Clint Ulahannan Thomas

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Professional Summary

Embedded Software Engineer with 6+ years of experience in automotive systems, vehicle communication, and networking platforms, specializing in C/C++ development, Ethernet-based gateway systems, and embedded Linux environments. Experienced in designing and integrating communication modules across CAN, CAN-FD, and Ethernet networks, with strong capabilities in system integration, debugging, and cross-layer issue resolution. Proven track record of delivering high-quality software for global OEM programs including Nissan, Subaru, and Toyota. Skilled in requirement analysis, validation, and cross-functional collaboration within safety-critical environments. PMP® certified with experience in structured development methodologies and system-level delivery.

Technical Skills

- **Programming & Software Development:** C, C++, Python, SQL (Basic)
 - **Embedded & Networking:** CAN, CAN-FD, Ethernet, MQTT, TCP/IP fundamentals, UDP, basic understanding of L2/L3 communication and VLAN concepts
 - **Operating Systems & Platforms:** Linux (CLI, debugging, log analysis), Windows
 - **Cybersecurity:** AES-128 Encryption Implementation
 - **AI / ML & Data Processing:** CNN, SVM, IoT Security, Sensor Data Processing
 - **Testing & Quality Engineering:** GTest, Test Automation, ISTQB Testing Practices
 - **Tools & Engineering Platforms:** Git, Jira, Confluence, DOORS, IMS, CANoe, Qt, GitHub Copilot, ChatGPT
 - **Project Management & Leadership:** Project Management Professional (PMP®), Predictive (Waterfall), Agile, Scrum, Requirement Engineering, Risk Management, Stakeholder Collaboration, Technical Mentorship
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Professional Experience

AUMOVIO India Private Limited (formerly Continental Automotive Components India Pvt. Ltd.)

- **Technical Specialist - HMI Developer | Feb 2026 - Apr 2026**
 - Completed training in automotive HMI and UX development with exposure to CGI-based interface design and visualization tools.
 - Gained understanding of user interface integration within automotive systems and infotainment environments.

AUMOVIO Autonomous Mobility India Private Limited (formerly Continental Automotive Mobility India)

- **Technical Specialist – Algorithm Developer (Radar Projects) | Jan 2023 – Jan 2026**
 - Designed object tracking algorithms for ADAS radar systems supporting OEM programs including Nissan, Subaru, and Toyota.
 - Implemented Time-to-Collision (TTC) and ID-consistency prioritization algorithms to improve object detection reliability.
 - Collaborated with system engineering, validation, and integration teams to support radar feature development.
 - Performed requirement engineering, documentation traceability, and unit testing using GTest.
 - Delivered internal training sessions on AI-assisted engineering tools (GitHub Copilot) to improve development productivity.

Continental Automotive Components (India) Pvt. Ltd.

- **Engineer – Gateway Application Software Developer | Apr 2020 – Dec 2022**

- Developed and integrated 100+ gateway communication modules across CAN, CAN-FD, and Ethernet networks.
- Enabled inter-ECU communication by implementing routing logic between heterogeneous vehicle networks.
- Performed system-level debugging across multiple layers including application, transport, and network layers.
- Built tools such as Canoe Broadcaster and Remote Hardware Controller to improve validation efficiency.
- TCP/IP (Ethernet Communication)
 - Worked with TCP/IP protocols as part of Ethernet-based vehicle communication.
 - Supported reliable (TCP) and real-time (UDP) communication between ECUs.
 - Analyzed packet flow, latency, and communication failures during debugging.
 - Enabled data exchange for diagnostics and service-oriented communication over IP networks.
- Ethernet Switch (Switched Network Handling)
 - Developed gateway software operating over switched Ethernet networks connecting multiple ECUs.
 - Ensured correct message routing across ECUs communicating through Ethernet switches.
 - Debugged end-to-end communication paths involving frame forwarding and MAC-level behavior.
- VLAN (Network Segmentation)
 - Worked with VLAN-based network architecture concepts used to separate ADAS, infotainment, and diagnostics traffic.
 - Ensured gateway routing logic aligned with segmented network design.
 - Supported debugging and integration in environments where communication depended on VLAN separation.
- Performed integration testing, validation, and root cause analysis for communication issues.
- Collaborated with cross-functional teams for requirement analysis and issue resolution.
- Developed automation tools for validation workflows and testing efficiency.
- **Intern – Algorithm Test Engineer (SIL Team) | Jul 2019 – Apr 2020**
 - Supported validation of emergency braking algorithms (EBA) in simulation environments.
 - Developed Python automation tools to analyze validation data and improve test efficiency.
 - Acted as Single Point of Contact (SPOC) for specific testing initiatives within the validation team.

Projects

- **Radar Object Prioritization Algorithm:** Developed radar object prioritization logic using Time-to-Collision and ID-consistency metrics to improve ADAS perception reliability.
- **Vehicle Gateway Communication Platform:** Developed embedded gateway modules supporting vehicle communication across CAN and Ethernet networks, enabling data routing and inter-ECU communication.
- **Canoe Broadcaster & Remote Controller:** Built a setup enabling remote monitoring and control of test hardware environments to improve validation workflows.
- **Engineering Automation Tools:** Developed Python utilities for validation data analysis and duplicate record detection.
- **AI-Based Intrusion Detection Prototype:** Built a machine learning prototype using MQTT and CNN models to detect anomalies in IoT network traffic.
- **Driver Drowsiness Detection System:** Implemented driver fatigue detection using image processing techniques and SVM classification models.
- **IoT Home Automation Platform:** Designed a cloud-connected automation platform using MQTT and Python.

Education

- **M.Tech – Computer Science & Engineering** | Vellore Institute of Technology | 2020 | CGPA: 9.01
- **B.E – Information Science & Engineering** | The Oxford College of Engineering | 2018 | 61.11%

Awards & Recognition

- **SPOT Award:** Timely delivery of AES-128 encryption modules.
- **Continental Idea Management (CIM) Award:** Innovation in automation tooling improving validation processes.
- **Continental Business Solution (CBS) Award:** Canoe Broadcaster & Remote Controller Project (2021).
- **Star of the Month:** Testing & Validation Excellence (Jan 2021).
- **Best Paper Award:** 17th International Conference on Science, Engineering, and Technology.
- **Best Project Award:** Final Year Engineering.
- **Piano Grade 6 & Music Theory Grade 5:** Associated Board of the Royal Schools of Music.

Leadership & Engagement

- **Technical Mentor:** Guided junior engineers and supported onboarding of new hires.
- **Townhall Host:** Facilitated internal leadership townhall sessions.
- **Team Engagement Initiatives:** Organized activities to strengthen team collaboration.

Certifications

- Project Management Professional (PMP®) — Project Management Institute
- ISTQB Foundation Level Certified Tester
- Certified Course in Spatial Data Science and Application — Coursera
- Certified NPTEL Course on Multicore Computer Architecture